SLO-04: SLO Alerting

> **Series:** SLO — Service Level Objectives | **Notebook:** 4 of 5 | **Created:** June 2026 | **Last Updated:** 08/28/2026

Overview

The highest-value alert a platform can send is "you are about to breach an SLO." This notebook covers **burn-rate alerting** — why it beats thresholding the raw SLI, how fast-burn and slow-burn multiwindow alerts work, how Dynatrace surfaces SLO alerts, and how to route them through Workflows without drowning the team in noise.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS Gen3 with the SLO app and AutomationEngine (Workflows)
Permissions	`settings:objects:read/write` , workflow authoring permissions

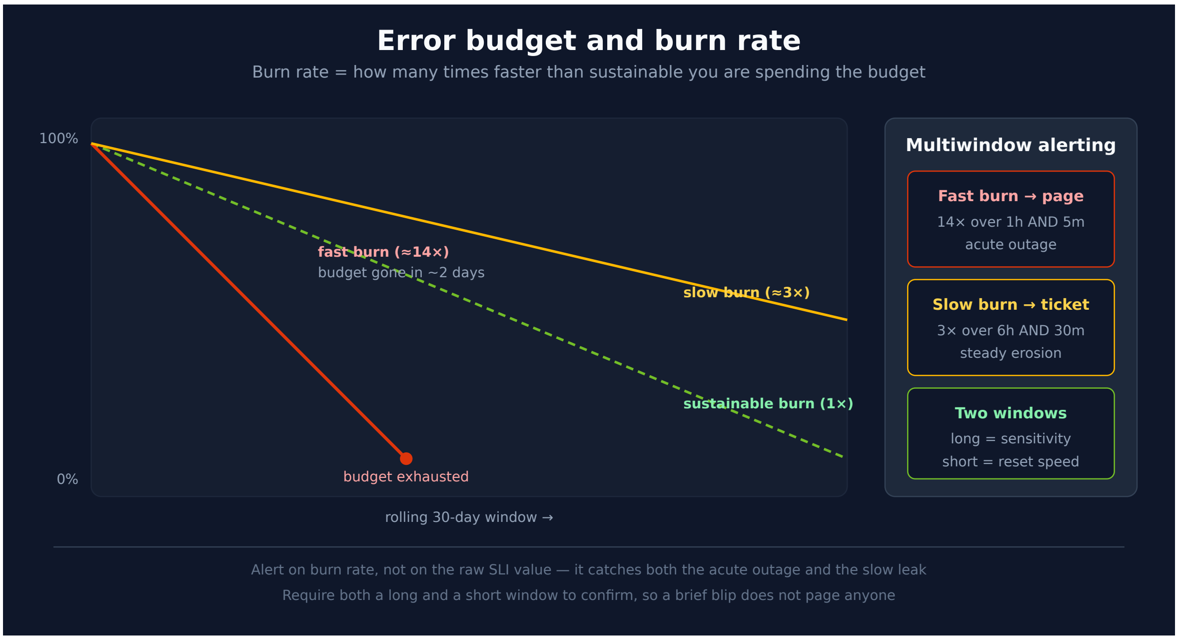
Requirement	Details
Prior reading	SLO-03 (error budgets and burn rate)

1. Why Burn-Rate, Not Threshold-on-SLI

The naive approach — alert when the SLI drops below the target — is a bad signal:

- **It fires late on slow leaks.** A rolling-30-day SLI barely moves when a service degrades for an hour, so the alert comes long after the damage.
- **It flaps on momentary dips.** A single bad interval can cross the line and recover, paging someone for nothing.

Burn rate fixes both. Because it measures the *rate* of budget consumption over a short window, a sharp degradation produces a high burn rate immediately, while a brief blip does not sustain. You alert on "the budget is draining too fast," which is what you actually care about.



2. Fast-Burn and Slow-Burn

The established SRE pattern uses two alert tiers, each requiring **two windows** to agree before firing:

Tier	Burn rate	Windows (long AND short)	Response
Fast burn	~14×	1 hour AND 5 minutes	Page — acute, budget gone in ~2 days
Slow burn	~3×	6 hours AND 30 minutes	Ticket — steady erosion

The **long window** gives sensitivity (it confirms the burn is real); the **short window** gives fast reset (it stops alerting quickly once the burn stops). Requiring both prevents a momentary spike from paging anyone. The query below evaluates a fast-burn condition; wire it into a detector (Section 3):

```
// Fast-burn check: is the 1h burn rate &gt;= 14x against a 99.5% target?
timeseries {
  total = sum(dt.service.request.count),
  failures = sum(dt.service.request.failure_count)
}, from:-1h, interval:5m
| fieldsAdd bad_ratio = failures[] / total[]
| fieldsAdd observed_bad = arrayAvg(bad_ratio)
| fieldsAdd burn_rate = observed_bad / (1.0 - 0.995)
| fieldsAdd fast_burn = if(burn_rate &gt;= 14, 1, else: 0)
| fields burn_rate, fast_burn
```

3. Configuring SLO Alerts in Dynatrace

Dynatrace surfaces SLO health as events you can alert on. In practice you have two routes:

- **Built-in SLO alerting** — the SLO definition itself can raise an alert when the error budget / burn rate crosses a configured level. This is the simplest path and keeps the alert tied to the SLO object.
- **Anomaly Detection on the burn-rate query** — when you want the full multiwindow fast/slow-burn logic, evaluate a burn-rate query in the Anomaly Detection app and let it raise a Davis event. See AIOPS-02 §4 for the detector build flow and event template.

Either way the breach becomes a Davis problem/event carrying the SLO context — which is what a workflow routes on.

The documented recipe for the burn-rate query is to append two lines to the SLI definition, then feed that into Anomaly Detection:

```
| fieldsAdd sli = "YOUR SLI"
| fieldsAdd target = "YOUR SLO-target" // in percentage
| fieldsAdd burnRate = ((100 - sli[]) / (100 - target))
| fieldsRemove sli
```

If you want one aggregated burn-rate value across every contributing entity rather than a per-entity series (at the cost of losing which entity is driving it), add a `summarize sloBurnRate = avg(burnRate[]), timeframe = takeFirst(timeframe), interval = takeFirst(interval)` step. Segments are **not currently supported** inside Anomaly Detection, so a segment-scoped SLO's burn-rate alert needs the segment expressed as a query filter instead.

Fast-burn tuning, per Dynatrace's own recommendation: Anomaly Detection's **-1h look-back window** is called out as well-suited for fast-burn alerting, with **10–14** as "a good starting point" for the static burn-rate threshold at that window — adjust up or down based on the SLO's criticality and evaluation period. Dynatrace also recommends four event properties on the custom alert so the resulting event carries enough context to route and correlate automatically: **`dt.source` entity** (attaches the affected service entities), **`event.type`** (`ERROR_EVENT` / `AVAILABILITY_EVENT` / `PERFORMANCE_EVENT` — matches the event into Dynatrace Intelligence RCA), **`slo.name`** (ties the event back to the SLO, which is unique per environment), and **`dt.owner`** (drives automatic routing to the right team).

> The **slow-burn tier** (longer look-back, lower threshold, ticket-not-page routing) below is the standard SRE multiwindow pattern layered on top of this recipe — Dynatrace's docs describe the fast-burn -1h/10–14 configuration explicitly but do not publish an equivalent numeric recommendation for a slow-burn window, so treat those values as a sound starting point to tune against your own traffic and SLO criticality rather than a vendor-specified default.

4. Routing SLO Breaches

An SLO breach raises a problem; routing it is the WFLOW series' job, not a separate mechanism:

1. **Problem-trigger workflow** filters for the SLO-breach event (by event properties — name, the owning team/zone you set in the detector's event template).
2. **Route to the owning team's channel** — Slack, Teams, PagerDuty, or an ITSM incident.

3. **Differentiate by tier** — fast-burn → page (PagerDuty / on-call); slow-burn → ticket (Jira / ServiceNow).

This is exactly the routing covered in WFLOW-04. The key dependency is upstream: the burn-rate detector's **event template must carry the team/service metadata** (AIOPS-02 §4) or the workflow has nothing to filter on.

5. Avoiding Alert Fatigue

- **Only page on fast burn.** Slow burn is a ticket, not a 3 a.m. call.
- **Require both windows.** The short window is what stops the alert from re-firing as conditions recover.
- **Set recovery/dealerting deliberately.** Use the detector's `dealertingSamples` (AIOPS-02 §4) so an alert closes cleanly rather than flapping.
- **One SLO breach, one notification.** Let the problem group related signals; do not also alert on the underlying raw metrics or you double-notify.
- **Review what fired.** If an SLO pages and no one acts, the target or the burn thresholds are wrong — tune them.

> **Sources:** Service-Level Objectives (DT docs) — burn-rate DQL append pattern, -1h/10–14 fast-burn recommendation, and the four recommended event properties are cited verbatim from the "Monitor error-budget burn rates" section (fetched 07/01/2026); Site Reliability Engineering — Alerting on SLOs (Google SRE Workbook). Fast-burn query re-executed against a live tenant 08/28/2026 (burn_rate 9.05, fast_burn 0 — below the 14x trigger, which is the expected shape for a healthy window). **Derived:** the two-tier fast/slow-burn window pairs are the standard SRE values layered onto the Dynatrace-documented fast-burn recipe above — Dynatrace's own docs only specify the fast-burn side.
